Requirements Gathering

UK Train Rides Analytics: Delays, Cancellations, Refunds and Revenue Forecasting (Power BI)

DEPI Graduation Project

Team: NHA-5-049

Project Leader: Ahmed Nasser Abdelmoghney Jafer

Date: October 2026

1. Problem Statement

A UK railway company sells tickets online and at stations. Some journeys are delayed or cancelled, which causes lost revenue, refund requests and unhappy customers. The company needs to know when and why these problems happen, how much revenue they affect, and which department can solve each one.

2. Stakeholders

- Operations Control: monitors delays and cancellations.
- Maintenance and Engineering: handles technical issues.
- Signalling and Infrastructure: handles signal failures.
- HR and Workforce Planning: handles staffing and staff shortages.
- Network Planning: handles timetables, weather planning and traffic.
- Customer Service: handles refund requests and customer communication.
- Management: uses the dashboard to make decisions.

3. Business Questions

- When do delays and cancellations happen, and how much revenue do they affect?
- Which causes (weather, signal failure, technical issue, staffing, staff shortage, traffic) affect revenue the most?
- Which routes have the highest cancellation rates?
- How do delays and cancellations differ between tickets bought online and at stations?
- How much refunded money comes from delayed trips compared with cancelled trips?
- Which customers had a delayed or cancelled trip but did not request a refund?
- What are the peak hours, and how do delays and cancellations behave during them?
- Which department can solve each cause, and how much revenue can be recovered?
- What will next month's rides, daily revenue and demand by ticket class look like?

4. The Five Problems and Their Owners

#	Problem	Owner
1	Delays and how to handle them	Aliaa Saeed Abd Elaziz Sarhan
2	Cancellations and how to handle them	Eman Ahmed Ebrahim Kenawy
3	Refund requests	Mohamed Baligh Mahmoud Ateya
4	Handled customers (delayed or cancelled trip, no refund requested)	Ahmed Nasser Abdelmoghney Jafer
5	Peak hours	Seba Ahmed Adel Yehia

5. Key Performance Indicators (KPIs)

The dashboard tracks the following indicators. All of them are calculated in Power BI using DAX measures.

KPI	Definition	Value in the Data
Total Trips	Number of ticket transactions	31,653
Total Revenue	Sum of ticket prices	£741,921
Delay Rate %	Delayed trips divided by total trips	7.2%
Cancellation Rate %	Cancelled trips divided by total trips	5.9%
Refund Rate %	Trips with a refund request divided by total trips	3.5%
Avg Delay Minutes	Average delay of delayed trips	42.2 minutes
Handled Customers	Delayed or cancelled trips with no refund request (assumed handled)	3,054
Handled Revenue	Revenue of handled customers	£133,568
Problem Revenue	Revenue of delayed and cancelled trips	£172,270
Recovered Revenue (Assumed)	Problem revenue multiplied by the assumed success rate of each cause's solution	About £46,210
Hour Share %	Share of tickets sold per departure hour	Varies by hour
Peak Flag	Peak if the hour share is above the threshold set by the Peak Multiplier	Hours 6-8 and 16-18 at a multiplier of 1.5

6. Functional Requirements

- FR1: The system shall load and clean the railway ticket dataset (31,653 transactions).
- FR2: The system shall store the data in a star schema with one fact table and dimension tables.
- FR3: The dashboard shall show delays, cancellations and refund requests by cause, route, ticket type, purchase method and time.
- FR4: The dashboard shall identify peak hours from the ticket data and allow the user to change the peak threshold.
- FR5: The dashboard shall show customers who had a delayed or cancelled trip and did not request a refund (handled customers).
- FR6: The dashboard shall link each cause to a responsible department, a proposed solution and an assumed success rate.
- FR7: The dashboard shall estimate recovered revenue from the proposed solutions.
- FR8: The dashboard shall forecast next month's rides, daily revenue and demand by ticket class.
- FR9: The dashboard shall support filtering by date, station, ticket class and journey status.

7. Non-Functional Requirements

- NFR1: The dashboard shall refresh without errors when the data source changes.
- NFR2: The data model shall follow a star schema to keep it fast and simple.
- NFR3: The dashboard shall be interactive and easy to read for non-technical users.
- NFR4: All assumed values (success rates, handled customers) shall be labelled as assumptions.
- NFR5: Documents and code shall be stored in the team GitHub repository.

8. Data Description

The dataset contains 31,653 railway ticket transactions for journeys between 1 January 2024 and 30 April 2024. Each row is one ticket purchase. The full data dictionary is included in this folder (railway_data_dictionary.csv).

8.1 Fields

Field	Description
Transaction ID	Unique identifier for a ticket purchase
Date of Purchase	Date the ticket was purchased
Time of Purchase	Time the ticket was purchased
Purchase Type	Online or at a station
Payment Method	Contactless, Credit Card or Debit Card
Railcard	Adult, Senior, Disabled or None (railcard holders get 1/3 off)
Ticket Class	Standard or First Class
Ticket Type	Advance (1/2 off), Off-Peak (1/4 off) or Anytime (full price)
Price	Final cost of the ticket
Departure Station	Station where the passenger boards
Arrival Destination	Station where the passenger exits
Date of Journey	Date the train departed
Departure Time	Time the train departed
Arrival Time	Scheduled arrival time (can be the next day)
Actual Arrival Time	Actual arrival time (empty for cancelled trips)
Journey Status	On Time, Delayed or Cancelled
Reason for Delay	Reason for the delay or cancellation
Refund Request	Whether the passenger requested a refund

8.2 Data Preparation

- Reason for Delay: inconsistent values were standardized (for example, "Weather Conditions" became "Weather" and "Signal failure" became "Signal Failure"), and empty values were replaced with "No Delay".

- Actual Arrival Time: empty values were kept for cancelled trips, because these trips never arrived.

- Dimension tables were created for purchase type, payment method, railcard, ticket class, ticket type, stations, journey status, refund request, reason, customer action and follow-up action.

- Customer Action and Follow-up Action are assumed actions derived from the journey status, refund request and reason. The dataset does not record how customers were handled.

- Peak hours detected from the ticket data (6-8 am and 4-6 pm) match the Off-Peak definition in the data dictionary.

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